

Healthcare Dataset Analysis

A Power BI Dashboard Report

1. Introduction

This report presents an analysis of a healthcare dataset, visualized through an interactive Power BI dashboard. The dataset captures patient-level information across several dimensions, including medical conditions, age groups, insurance providers, admission types, billing amounts, and length of hospital stay. The objective of this analysis is to understand patterns in patient distribution, healthcare costs, and hospital utilization in order to support data-driven insights into healthcare delivery and resource planning.

The dashboard consolidates key performance indicators (KPIs) such as total patients, total billing amount, average billing amount, and average length of stay, alongside visual breakdowns by medical condition, age group, and admission trends over time. Interactive filters allow the data to be sliced by insurance provider (Aetna, Blue Cross, Cigna, Medicare) and admission type (Elective, Emergency, Urgent), enabling deeper, more targeted exploration of the dataset.

2. Purpose of this Report

The purpose of this report is to document the structure and key findings of the Healthcare Dataset Analysis dashboard. The following pages present the dashboard visual itself, followed by a detailed explanation of each of its components, the metrics they display, and the insights that can be drawn from them.

3. Dashboard Overview

The image below shows the full Power BI report as built in the Power BI Desktop environment, including the ribbon, filter panel, and the visualizations on the report canvas.

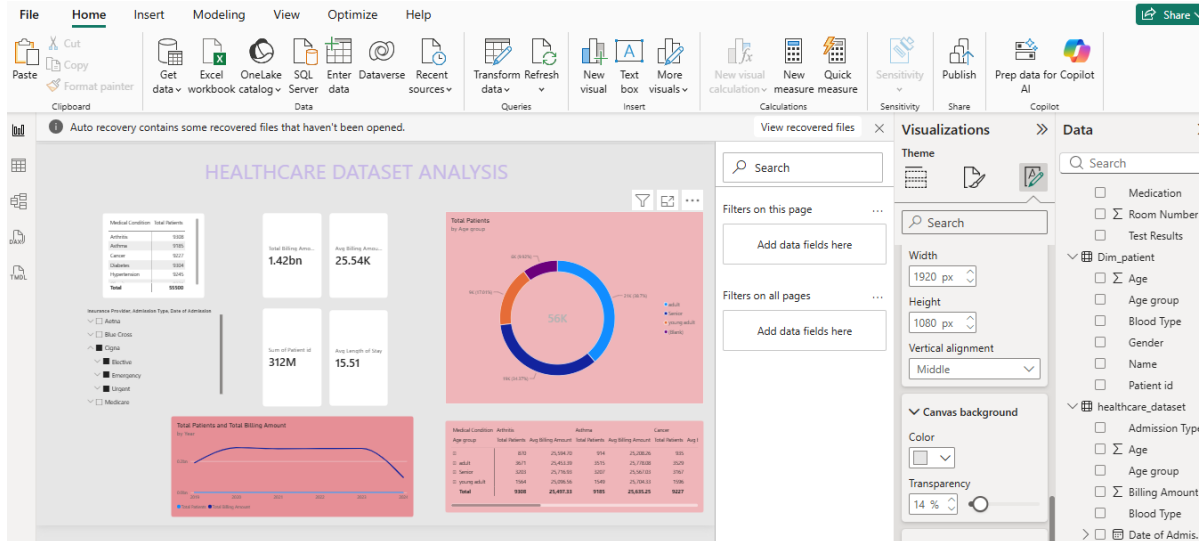


Figure 1: Healthcare Dataset Analysis — Power BI Dashboard

4. Explanation of the Dashboard

4.1 Key Performance Indicator (KPI) Cards

At the top-left of the canvas, four KPI cards summarize the dataset at a glance:

- **Total Billing Amount:** 1.42bn — the combined billing value across all patient records.
- **Average Billing Amount:** 25.54K — the mean billing amount per patient.
- **Sum of Patient ID:** 312M — an aggregate of patient ID values, generally used here as a record-count style measure rather than a meaningful sum.
- **Average Length of Stay:** 15.51 — the average number of days patients remain admitted.

4.2 Medical Condition Summary Table

A table on the left lists total patients by medical condition — Arthritis, Asthma, Cancer, Diabetes, and Hypertension — with a grand total row of approximately 55,500 patients. Each condition contributes a broadly similar share, indicating a fairly evenly distributed patient population across the five tracked conditions.

4.3 Total Patients by Age Group (Donut Chart)

The donut chart breaks down the total patient count (56K) by age group — Adult, Senior, and Young Adult — with the Adult segment forming the largest share of the circle, followed by Senior and Young Adult. This highlights that the middle-aged adult population accounts for the bulk of hospital admissions in this dataset.

4.4 Total Patients and Total Billing Amount Over Time (Line Chart)

A combination line chart tracks total patients and total billing amount across admission years. The two lines move in a closely related pattern, rising toward the middle years before tapering off, suggesting that billing amount scales roughly in proportion to patient volume over time rather than being driven by isolated high-cost cases.

4.5 Age Group vs. Medical Condition Matrix

The table on the right cross-tabulates total patients and average billing amount for each age group (Adult, Senior, Young Adult) against selected medical conditions (Arthritis, Asthma, Cancer). This view is useful for identifying which age groups incur higher average costs for specific conditions, supporting more targeted cost and care analysis.

4.6 Filters Panel

On the far left, a slicer panel lists Insurance Provider (Aetna, Blue Cross, Cigna, Medicare) and, nested under each provider, Admission Type (Elective, Emergency, Urgent). These filters let a user interactively narrow the entire report to a specific insurer and admission type, allowing comparisons

such as how emergency admissions differ in cost or volume across insurance providers.

5. Conclusion

Overall, the dashboard provides a compact but comprehensive view of the healthcare dataset, combining high-level KPIs with condition-based, age-based, and time-based breakdowns. The interactive filters further allow the data to be explored by insurance provider and admission type, making the report a useful tool for identifying cost drivers, patient volume trends, and utilization patterns within the dataset.